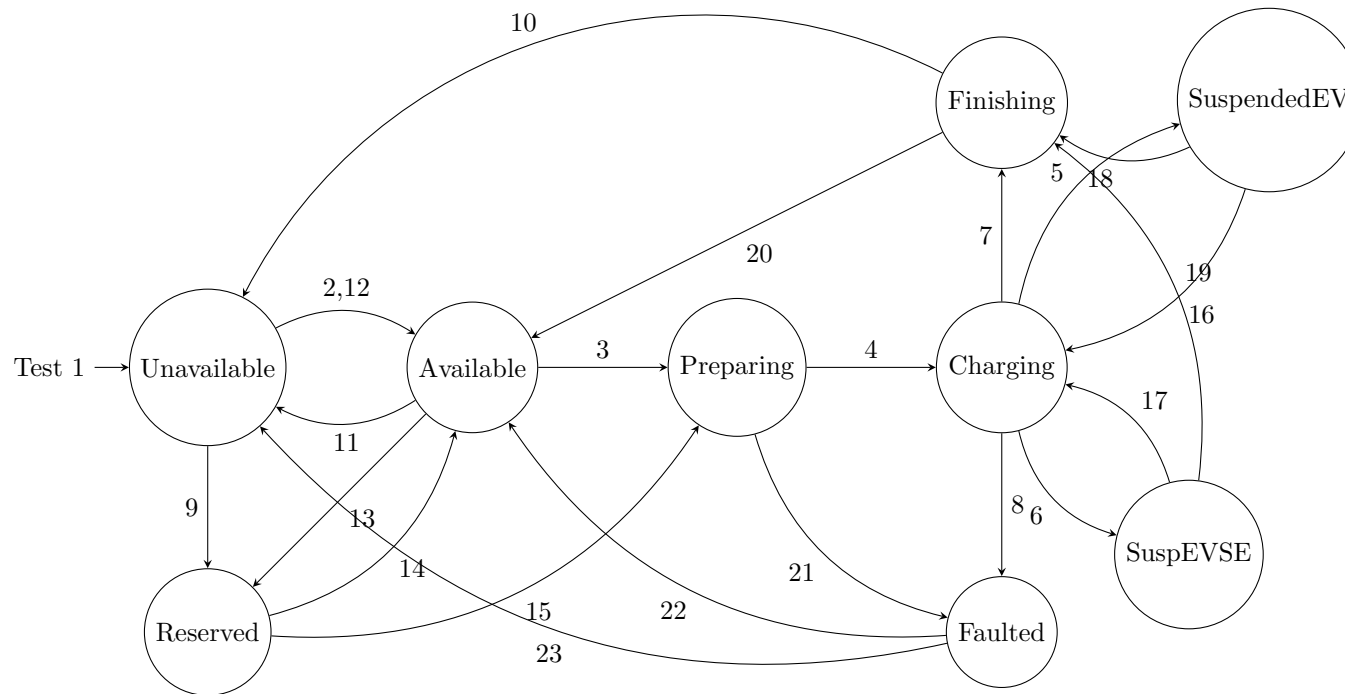


Evse.py Unit Test Report

MEA OCPP Certification (Automated)

January 15, 2026

State Transition Diagram



Unit Test Execution Results

ID	Test Case	State Check	Log Preview / Trace	Result
1	test_initial_state	Logic Check	WHITE-beet-EI firmware version: 1.0.0 RFIDSim stopped.	PASSED
2	test_transition_to_available	Logic Check	WHITE-beet-EI firmware version: 1.0.0 Set the CP mode to EVSE Set the CP duty cycle to 100% Start the CP service Start SLAC in EVSE mode RFIDSim started. Call simulate_scan(id.tag) to trigger. State transition: Unavailable -> Available	PASSED
3	test_transition_to_preparing	Logic Check	WHITE-beet-EI firmware version: 1.0.0 RFIDSim started. Call simulate_scan(id.tag) to trigger. RFID Scanned: RFID.TAG EVSE Authorized by OCPP: RFID.TAG EV already connected, transitioning to Preparing for authorized tag: RFID.TAG State transition: Available -> Preparing EV already connected	PASSED
4	test_transition_to_charging	Logic Check	WHITE-beet-EI firmware version: 1.0.0 "Energy transfer mode selected" received State transition: Preparing -> Charging Maximum voltage: 400 Maximum current: 32 RFIDSim stopped.	PASSED
5	test_transition_to_suspended_ev	Logic Check	WHITE-beet-EI firmware version: 1.0.0 "DC Charge Parameters Changed" received EV maximum current: 32A EV maximum voltage: 400V EV ready: True Error code: 0 SOC: 50% EV target current: 0A State transition: Charging -> SuspendedEV RFIDSim stopped.	PASSED

ID	Test Case	State Check	Log Preview / Trace	Result
6	test_transition_to_suspended_evse	Logic Check	WHITE-beet-EI firmware version: 1.0.0 "DC Charge Parameters Changed" received EV maximum current: 32A EV maximum voltage: 400V EV ready: True Error code: 0 SOC: 50% EV target current: 10A State transition: Charging -> SuspendedEVSE RFIDSim stopped.	PASSED
7	test_transition_to_finishing	Logic Check	WHITE-beet-EI firmware version: 1.0.0 "Session stopped" received State transition: Charging -> Finishing Closure type: 0 RFIDSim stopped.	PASSED
8	test_transition_to_faulted	Logic Check	WHITE-beet-EI firmware version: 1.0.0 "Session Error" received State transition: Charging -> Faulted Session error: 0: Unspecified RFIDSim stopped.	PASSED
9	test_transition_to_reserved	Logic Check	WHITE-beet-EI firmware version: 1.0.0 State transition: Unavailable -> Reserved Reservation set until 2026-01-01 12:00:00+00:00 RFIDSim stopped.	PASSED
10	test_transition_to_unavailable_via_pending	Logic Check	WHITE-beet-EI firmware version: 1.0.0 Scheduling availability change to Inoperative State transition: Finishing -> Available Pending 'Inoperative' availability detected, transitioning to Unavailable. State transition: Available -> Unavailable RFIDSim stopped.	PASSED
11	test_transition_available_to_unavailable	Logic Check	WHITE-beet-EI firmware version: 1.0.0 State transition: Available -> Unavailable RFIDSim stopped.	PASSED

ID	Test Case	State Check	Log Preview / Trace	Result
12	test_transition_unavailable_to_available	Logic Check	WHITE-beet-EI firmware version: 1.0.0 State transition: Unavailable -> Available RFIDSim stopped. ===== 23 passed in 0.04s ===== RFIDSim stopped. RFIDSim stopped.	PASSED
13	test_transition_available_to_reserved	Logic Check	WHITE-beet-EI firmware version: 1.0.0 State transition: Available -> Reserved RFIDSim stopped.	PASSED
14	test_transition_reserved_to_available	Logic Check	WHITE-beet-EI firmware version: 1.0.0 State transition: Reserved -> Available RFIDSim stopped.	PASSED
15	test_transition_reserved_to_preparing	Logic Check	WHITE-beet-EI firmware version: 1.0.0 EV already connected State transition: Reserved -> Preparing RFIDSim stopped.	PASSED
16	test_transition_suspendedev_to_charging	Logic Check	WHITE-beet-EI firmware version: 1.0.0 "DC Charge Parameters Changed" received EV maximum current: 32A EV maximum voltage: 400V EV ready: True Error code: 0 SOC: 50% EV target current: 10A State transition: SuspendedEV -> Charging RFIDSim stopped.	PASSED

ID	Test Case	State Check	Log Preview / Trace	Result
17	test_transition_suspendedevse_to_charging	Logic Check	WHITE-beet-EI firmware version: 1.0.0 "DC Charge Parameters Changed" received EV maximum current: 32A EV maximum voltage: 400V EV ready: True Error code: 0 SOC: 50% EV target current: 10A State transition: SuspendedEVSE -> Charging RFIDSim stopped.	PASSED
18	test_transition_suspendedev_to_finishing	Logic Check	WHITE-beet-EI firmware version: 1.0.0 State transition: SuspendedEV -> Finishing RFIDSim stopped.	PASSED
19	test_transition_suspendedevse_to_finishing	Logic Check	WHITE-beet-EI firmware version: 1.0.0 State transition: SuspendedEVSE -> Finishing RFIDSim stopped.	PASSED
20	test_transition_finishing_to_available	Logic Check	WHITE-beet-EI firmware version: 1.0.0 State transition: Finishing -> Available RFIDSim stopped.	PASSED
21	test_transition_preparing_to_faulted	Logic Check	WHITE-beet-EI firmware version: 1.0.0 "Session Error" received State transition: Preparing -> Faulted Session error: 0: Unspecified RFIDSim stopped.	PASSED
22	test_transition_faulted_to_available	Logic Check	WHITE-beet-EI firmware version: 1.0.0 Resetting EVSE... State transition: Faulted -> Available RFIDSim stopped.	PASSED
23	test_transition_faulted_to_unavailable	Logic Check	WHITE-beet-EI firmware version: 1.0.0 State transition: Faulted -> Unavailable RFIDSim stopped.	PASSED

A Raw Unit Test Logs

```

1 ===== test session starts =====
2 platform darwin -- Python 3.14.0, pytest-9.0.2, pluggy-1.6.0 -- /Users/rus/Development/MEA-V2G/.venv/bin/python
3 cachedir: .pytest_cache
4 rootdir: /Users/rus/Development/MEA-V2G
5 configfile: pytest.ini
6 collecting ... collected 23 items
7
8 tests/unit/test_evse_states.py::TestEvseStates::test_initial_state PASSED [ 4%]
9 tests/unit/test_evse_states.py::TestEvseStates::test_transition_available_to_reserved PASSED [ 8%]
10 tests/unit/test_evse_states.py::TestEvseStates::test_transition_available_to_unavailable PASSED [ 13%]
11 tests/unit/test_evse_states.py::TestEvseStates::test_transition_faulted_to_available PASSED [ 17%]
12 tests/unit/test_evse_states.py::TestEvseStates::test_transition_faulted_to_unavailable PASSED [ 21%]
13 tests/unit/test_evse_states.py::TestEvseStates::test_transition_finishing_to_available PASSED [ 26%]
14 tests/unit/test_evse_states.py::TestEvseStates::test_transition_preparing_to_faulted PASSED [ 30%]
15 tests/unit/test_evse_states.py::TestEvseStates::test_transition_reserved_to_available PASSED [ 34%]
16 tests/unit/test_evse_states.py::TestEvseStates::test_transition_reserved_to_preparing PASSED [ 39%]
17 tests/unit/test_evse_states.py::TestEvseStates::test_transition_suspendedev_to_charging PASSED [ 43%]
18 tests/unit/test_evse_states.py::TestEvseStates::test_transition_suspendedev_to_finishing PASSED [ 47%]
19 tests/unit/test_evse_states.py::TestEvseStates::test_transition_suspendedevse_to_charging PASSED [ 52%]
20 tests/unit/test_evse_states.py::TestEvseStates::test_transition_suspendedevse_to_finishing PASSED [ 56%]
21 tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_available PASSED [ 60%]
22 tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_charging PASSED [ 65%]
23 tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_faulted PASSED [ 69%]
24 tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_finishing PASSED [ 73%]
25 tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_preparing PASSED [ 78%]
26 tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_reserved PASSED [ 82%]
27 tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_suspended_ev PASSED [ 86%]
28 tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_suspended_evse PASSED [ 91%]
29 tests/unit/test_evse_states.py::TestEvseStates::test_transition_to_unavailable_via_pending PASSED [ 95%]
30 tests/unit/test_evse_states.py::TestEvseStates::test_transition_unavailable_to_available PASSED [100%]
31
32 ===== PASSES =====
33 ----- TestEvseStates.test_initial_state -----
34 ----- Captured stdout call -----
35 WHITE-beet-EI firmware version: 1.0.0
36 ----- Captured stdout teardown -----
37 RFIDSim stopped.
38 ----- TestEvseStates.test_transition_available_to_reserved -----
39 ----- Captured stdout call -----
40 WHITE-beet-EI firmware version: 1.0.0
41 State transition: Available -> Reserved

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42 ----- Captured stdout teardown -----
43 RFIDSim stopped.
44 ----- TestEvseStates.test_transition_available_to_unavailable -----
45 ----- Captured stdout call -----
46 WHITE-beet-EI firmware version: 1.0.0
47 State transition: Available -> Unavailable
48 ----- Captured stdout teardown -----
49 RFIDSim stopped.
50 ----- TestEvseStates.test_transition_faulted_to_available -----
51 ----- Captured stdout call -----
52 WHITE-beet-EI firmware version: 1.0.0
53 Resetting EVSE...
54 State transition: Faulted -> Available
55 ----- Captured stdout teardown -----
56 RFIDSim stopped.
57 ----- TestEvseStates.test_transition_faulted_to_unavailable -----
58 ----- Captured stdout call -----
59 WHITE-beet-EI firmware version: 1.0.0
60 State transition: Faulted -> Unavailable
61 ----- Captured stdout teardown -----
62 RFIDSim stopped.
63 ----- TestEvseStates.test_transition_finishing_to_available -----
64 ----- Captured stdout call -----
65 WHITE-beet-EI firmware version: 1.0.0
66 State transition: Finishing -> Available
67 ----- Captured stdout teardown -----
68 RFIDSim stopped.
69 ----- TestEvseStates.test_transition_preparing_to_faulted -----
70 ----- Captured stdout call -----
71 WHITE-beet-EI firmware version: 1.0.0
72 "Session Error" received
73 State transition: Preparing -> Faulted
74 Session error: 0: Unspecified
75 ----- Captured stdout teardown -----
76 RFIDSim stopped.
77 ----- TestEvseStates.test_transition_reserved_to_available -----
78 ----- Captured stdout call -----
79 WHITE-beet-EI firmware version: 1.0.0
80 State transition: Reserved -> Available
81 ----- Captured stdout teardown -----
82 RFIDSim stopped.
83 ----- TestEvseStates.test_transition_reserved_to_preparing -----
84 ----- Captured stdout call -----
85 WHITE-beet-EI firmware version: 1.0.0
```

```
86 EV already connected
87 State transition: Reserved -> Preparing
88 ----- Captured stdout teardown -----
89 RFIDSim stopped.
90 ----- TestEvseStates.test_transition_suspendedev_to_charging -----
91 ----- Captured stdout call -----
92 WHITE-beet-EI firmware version: 1.0.0
93 "DC Charge Parameters Changed" received
94 EV maximum current: 32A
95 EV maximum voltage: 400V
96 EV ready: True
97 Error code: 0
98 SOC: 50%
99 EV target current: 10A
100 State transition: SuspendedEV -> Charging
101 ----- Captured stdout teardown -----
102 RFIDSim stopped.
103 ----- TestEvseStates.test_transition_suspendedev_to_finishing -----
104 ----- Captured stdout call -----
105 WHITE-beet-EI firmware version: 1.0.0
106 State transition: SuspendedEV -> Finishing
107 ----- Captured stdout teardown -----
108 RFIDSim stopped.
109 ----- TestEvseStates.test_transition_suspendedevse_to_charging -----
110 ----- Captured stdout call -----
111 WHITE-beet-EI firmware version: 1.0.0
112 "DC Charge Parameters Changed" received
113 EV maximum current: 32A
114 EV maximum voltage: 400V
115 EV ready: True
116 Error code: 0
117 SOC: 50%
118 EV target current: 10A
119 State transition: SuspendedEVSE -> Charging
120 ----- Captured stdout teardown -----
121 RFIDSim stopped.
122 ----- TestEvseStates.test_transition_suspendedevse_to_finishing -----
123 ----- Captured stdout call -----
124 WHITE-beet-EI firmware version: 1.0.0
125 State transition: SuspendedEVSE -> Finishing
126 ----- Captured stdout teardown -----
127 RFIDSim stopped.
128 ----- TestEvseStates.test_transition_to_available -----
129 ----- Captured stdout call -----
```



```
130 WHITE-beet-EI firmware version: 1.0.0
131 Set the CP mode to EVSE
132 Set the CP duty cycle to 100%
133 Start the CP service
134 Start SLAC in EVSE mode
135 RFIDSim started. Call simulate_scan(id_tag) to trigger.
136 State transition: Unavailable -> Available
137 ----- TestEvseStates.test_transition_to_charging -----
138 ----- Captured stdout call -----
139 WHITE-beet-EI firmware version: 1.0.0
140 "Energy transfer mode selected" received
141 State transition: Preparing -> Charging
142 Maximum voltage: 400
143 Maximum current: 32
144 ----- Captured stdout teardown -----
145 RFIDSim stopped.
146 ----- TestEvseStates.test_transition_to_faulted -----
147 ----- Captured stdout call -----
148 WHITE-beet-EI firmware version: 1.0.0
149 "Session Error" received
150 State transition: Charging -> Faulted
151 Session error: 0: Unspecified
152 ----- Captured stdout teardown -----
153 RFIDSim stopped.
154 ----- TestEvseStates.test_transition_to_finishing -----
155 ----- Captured stdout call -----
156 WHITE-beet-EI firmware version: 1.0.0
157 "Session stopped" received
158 State transition: Charging -> Finishing
159 Closure type: 0
160 ----- Captured stdout teardown -----
161 RFIDSim stopped.
162 ----- TestEvseStates.test_transition_to_preparing -----
163 ----- Captured stdout call -----
164 WHITE-beet-EI firmware version: 1.0.0
165 RFIDSim started. Call simulate_scan(id_tag) to trigger.
166 RFID Scanned: RFID_TAG
167 EVSE Authorized by OCPP: RFID_TAG
168 EV already connected, transitioning to Preparing for authorized tag: RFID_TAG
169 State transition: Available -> Preparing
170 EV already connected
171 ----- TestEvseStates.test_transition_to_reserved -----
172 ----- Captured stdout call -----
173 WHITE-beet-EI firmware version: 1.0.0
```

```
174 State transition: Unavailable -> Reserved
175 Reservation set until 2026-01-01 12:00:00+00:00
176 ----- Captured stdout teardown -----
177 RFIDSim stopped.
178 ----- TestEvseStates.test_transition_to_suspended_ev -----
179 ----- Captured stdout call -----
180 WHITE-beet-EI firmware version: 1.0.0
181 "DC Charge Parameters Changed" received
182 EV maximum current: 32A
183 EV maximum voltage: 400V
184 EV ready: True
185 Error code: 0
186 SOC: 50%
187 EV target current: 0A
188 State transition: Charging -> SuspendedEV
189 ----- Captured stdout teardown -----
190 RFIDSim stopped.
191 ----- TestEvseStates.test_transition_to_suspended_evse -----
192 ----- Captured stdout call -----
193 WHITE-beet-EI firmware version: 1.0.0
194 "DC Charge Parameters Changed" received
195 EV maximum current: 32A
196 EV maximum voltage: 400V
197 EV ready: True
198 Error code: 0
199 SOC: 50%
200 EV target current: 10A
201 State transition: Charging -> SuspendedEVSE
202 ----- Captured stdout teardown -----
203 RFIDSim stopped.
204 ----- TestEvseStates.test_transition_to_unavailable_via_pending -----
205 ----- Captured stdout call -----
206 WHITE-beet-EI firmware version: 1.0.0
207 Scheduling availability change to Inoperative
208 State transition: Finishing -> Available
209 Pending 'Inoperative' availability detected, transitioning to Unavailable.
210 State transition: Available -> Unavailable
211 ----- Captured stdout teardown -----
212 RFIDSim stopped.
213 ----- TestEvseStates.test_transition_unavailable_to_available -----
214 ----- Captured stdout call -----
215 WHITE-beet-EI firmware version: 1.0.0
216 State transition: Unavailable -> Available
217 ----- Captured stdout teardown -----
```

```
218 RFIDSim stopped.  
219 ===== 23 passed in 0.04s =====  
220 RFIDSim stopped.  
221 RFIDSim stopped.
```